STUDENT PERFORMANCE ANALYTICS AND AI-BASED PREDICTION

A Data Analytics and Machine Learning Project

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Python • Pandas • Matplotlib • Scikit-learn

Project report for educational demonstration

1. Abstract

This project demonstrates a basic data analytics and machine-learning workflow for estimating student final scores from study hours, attendance, assignment completion, quiz averages, and sleep hours. A synthetic dataset of 40 records is used to explore patterns, visualise variables, train a Random Forest regression model, and evaluate its test-set predictions.

2. Objectives

• Analyse student performance indicators using descriptive statistics.

• Visualise the relationship between study hours and final scores.

• Explore correlations among numerical variables.

• Train a Random Forest regression model to estimate final scores.

• Evaluate the model with MAE and R-squared (R^2).

3. Tools and Technologies

Python is used as the programming language. Pandas handles tabular data, Matplotlib creates visualisations, and Scikit-learn provides the train/test split, Random Forest regression algorithm, and evaluation metrics.

4. Dataset Description

Column	Description
student_id	Synthetic identifier for each sample
study_hours_per_day	Study time per day, in hours
attendance_percent	Attendance percentage
assignments_completed_out_of_10	Assignment completion score from 1 to 10
quiz_average_out_of_10	Quiz average from 1 to 10
sleep_hours_per_day	Sleep time per day, in hours
final_score	Synthetic final score from 20 to 100

The dataset is synthetic and does not contain information about real students.

5. Methodology

• Load and validate the CSV dataset.

• Inspect summary statistics and create exploratory charts.

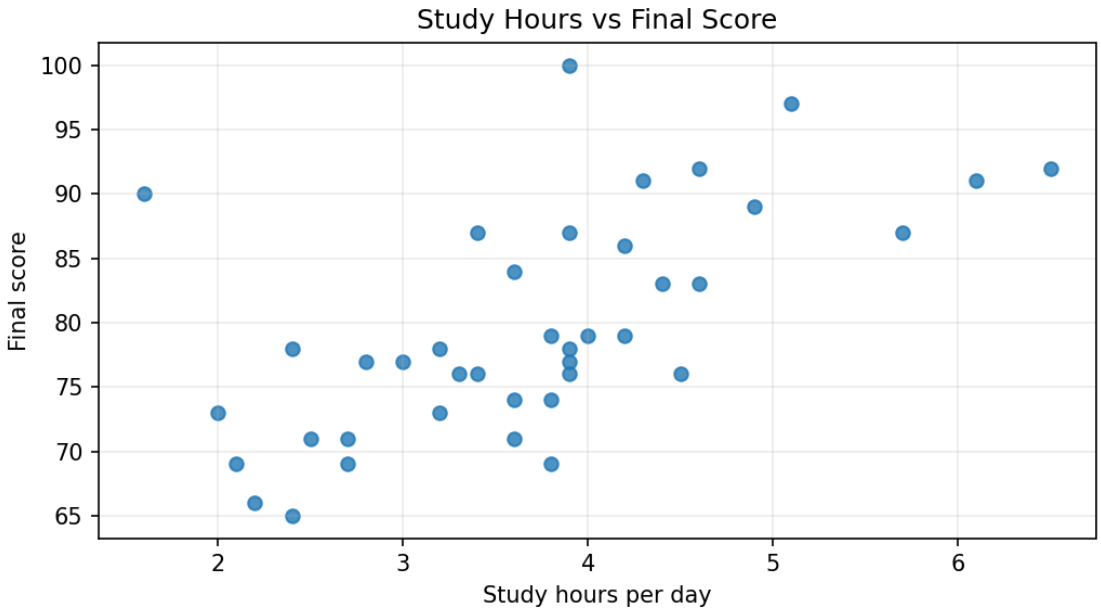
• Use five input features: study hours, attendance, assignments, quiz average, and sleep hours.

• Split the data into 75% training records and 25% test records using random_state=42.

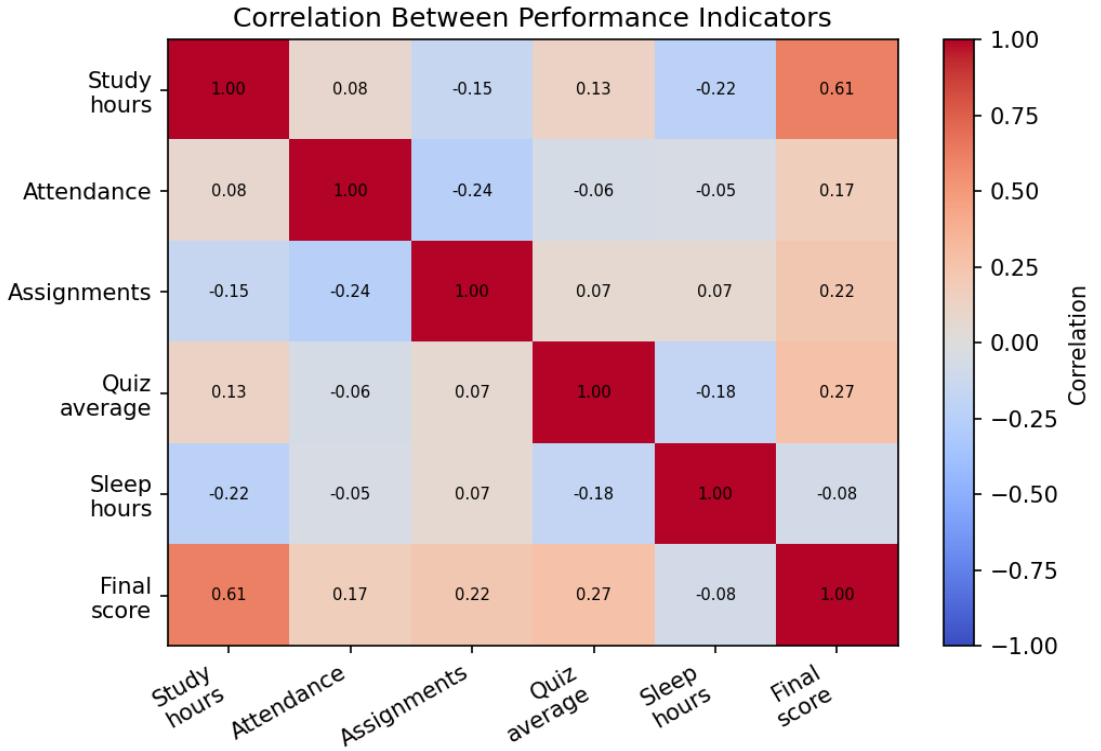
• Fit a RandomForestRegressor with 200 trees and evaluate predictions on the held-out test set.

6. Data Visualisation

The scatter plot below provides a first look at the relationship between study time and the generated final score. It is exploratory and should not be interpreted as proof of causation.



The correlation heatmap summarises pairwise linear relationships in the sample dataset. Correlations in synthetic data may reflect the way the sample was generated.



7. Model and Evaluation Metrics

Random Forest Regression: This ensemble method fits multiple decision trees and combines their predictions to estimate a numerical target such as a score.

Mean Absolute Error (MAE): The average absolute difference between the actual and predicted test scores. Lower values indicate smaller errors on this test split.

R-squared (R^2): A measure of how well the predictions account for variation in the test targets. On a very small test set, it can be unstable and may be negative.

The Python script calculates the actual MAE and R^2 when run. Metrics are deliberately not presented as fixed report claims because they depend on the dataset and model execution.

8. How to Install and Run

1. Install Python 3.10 or newer.
2. Open a terminal in the project folder.
3. Run `python -m pip install -r requirements.txt`.
4. Run `python student_performance_ai_project.py`.
5. Check the generated `output` folder for charts, model metrics, and test predictions.

9. Limitations and Ethical Considerations

• Only 40 synthetic records are included; this is too small to establish reliable generalisation.

• The train/test split leaves only 10 records for evaluation, so metrics can vary substantially.

• Synthetic patterns may not reflect real student behaviour or educational contexts.

• The model must not be used for real grading, admissions, discipline, or other consequential academic decisions.

• Real-world use would require appropriate consent, representative data, bias checks, privacy safeguards, validation, and human oversight.

10. Conclusion

This project illustrates an end-to-end introductory workflow for student-performance analytics: data preparation, visualisation, regression modelling, and evaluation. It is useful as a learning exercise, but its synthetic dataset and small sample size mean that its predictions are demonstration outputs only.

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